BUSINESS REQUIREMENTS DOCUMENT

(BRD)

AutoCare 360 – Universal Vehicle Maintenance Record System

Document Title	BRD- AutoCare 360 Universal Vehicle Maintenance Record System
Version	1.0
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1.Purpose

The purpose of this Business Requirements Document (BRD) is to define the business needs, objectives, scope, stakeholders and requirements for developing Auto Care 360 a universal digital vehicle maintenance record platform.

The system aims to replace traditional paper-based vehicle service records with a secure digital vehicle passport that stores complete maintenance history and allows authorized sharing between vehicle owners, garages, insurance companies and other stakeholders.

2.Executive Summary

Currently, vehicle maintenance records are mostly maintained manually through paper invoices, garage databases or disconnected systems. This creates difficulties in tracking vehicle history, verifying repairs, managing maintenance costs and transferring records during vehicle ownership changes.

Auto Care 360 introduces a centralized digital platform where every vehicle has a lifetime maintenance record. Vehicle owners maintain control over their data and can securely share selected information with authorized parties.

The platform supports:

- Vehicle owners
- Vehicle garages
- Fleet operators
- Insurance companies
- Leasing companies
- Vehicle buyers

The solution improves transparency, reduces fraud and creates a trusted digital vehicle service passport.

3.Business Problem

The absence of a centralized vehicle maintenance record system creates inefficiencies for vehicle owners, garages and related organizations.

A solution is required to,

- Digitize vehicle maintenance records
- Provide secure data ownership

- Enable authorized information sharing
- Improve vehicle resale transparency
- Reduce maintenance-related fraud

4.Objectives

Objective ID	Objective
OBJ-01	Create a centralized digital vehicle maintenance record platform
OBJ-02	Allow vehicle owners to manage lifetime vehicle history
OBJ-03	Enable garages to update verified service records
OBJ-04	Provide secure record-sharing capabilities
OBJ-05	Improve transparency in vehicle buying and selling
OBJ-06	Provide AI-based maintenance recommendations
OBJ-07	Support analytics-driven vehicle management

5.In Scope

Vehicle Management	• Vehicle registration • Vehicle profile management • Owner information management • Vehicle identification details
Maintenance Record Management	• Service history • Repair details • Parts replacement records • Cost tracking • Invoice uploads • Service images
Garage Portal	• Garage registration • Service record updates • Customer vehicle management • Invoice generation
Owner Portal	• Maintenance tracking • Document storage • Reminder management • Record sharing
Secure Sharing	• Secure links • QR code sharing • Expiry-based access • Permission-based viewing
Analytics Dashboard	• Secure links • QR code sharing • Expiry-based access

	• Permission-based viewing
AI Features	• Predictive maintenance • Invoice OCR scanning • AI vehicle assistant

6.Out of Scope

- Vehicle repair booking system
- Online spare parts marketplace
- Direct payment processing
- Vehicle manufacturing integration
- Real-time vehicle tracking

7.Business Requirements

ID	Requirement	Description
BR-01	Digital Vehicle Record Management	The system shall provide a digital profile for every registered vehicle. Vehicle details: • Registration number • VIN/Chassis number • Engine number • Vehicle type • Manufacturer • Model • Manufacturing year • Mileage • Owner information
BR-02	Complete Maintenance History	The system shall maintain complete service records. Each service record shall contain: • Service date • Garage details • Technician details • Maintenance category • Work performed

		• Parts replaced • Service cost • Invoice documents • Service images
BR-03	Garage Service Management	The system shall allow verified garages to: • Register their business • Manage customer vehicles • Add maintenance records • Generate quotations • Create invoices
BR-04	Vehicle Owner Control	The system shall allow vehicle owners to: • View maintenance history • Track expenses • Upload vehicle documents • Receive reminders • Transfer vehicle records
BR-05	Secure Data Sharing	Vehicle owners shall be able to share records securely. Sharing methods: 1. Secure Link Users can define: • Access duration • Recipient • Data visibility 2. QR Code Used for: • Vehicle sales • Insurance inspection • Vehicle verification
BR-06	Access Permission Management	Owners can select:

		1. Full Access Includes: • Complete service history • Costs • Documents 2. Limited Access Includes: • Service dates • Mileage • Garage details ❖ Selected Records Example: Allow: • Engine repair history Hide: • Insurance documents
BR-07	Data Protection Requirements	The system shall provide: 1. Protected Viewing Mode Restrictions: • No downloading • No printing • No copying • No exporting 2. Watermark Protection Shared records shall display: • Recipient name • Access ID • Date and time • Expiry date

BR-08	Audit Trail Management	The system shall record: • Who accessed records? • Access time • Device information • Sharing status
BR-09	AI Vehicle Health Assessment	The system shall generate: 1.Vehicle Health Passport Including: • Maintenance score • Mileage verification • Repair history • Service consistency Example: Vehicle Trust Score: 92/100

8.Functional Requirements

ID	Requirement
FR-01	User registration and authentication
FR-02	Vehicle profile creation
FR-03	Service record creation
FR-04	Document upload
FR-05	Garage verification
FR-06	Reminder notification
FR-07	Secure record sharing
FR-08	QR code generation
FR-09	Analytics dashboard
FR-10	AI maintenance prediction

9. Non-Functional Requirements

Category	Requirement
Security	User authentication and access control
Performance	Fast record retrieval
Availability	24/7 system availability
Scalability	Support large number of vehicles
Usability	Easy mobile and web experience
Reliability	Prevent data loss
Privacy	Owner-controlled data sharing

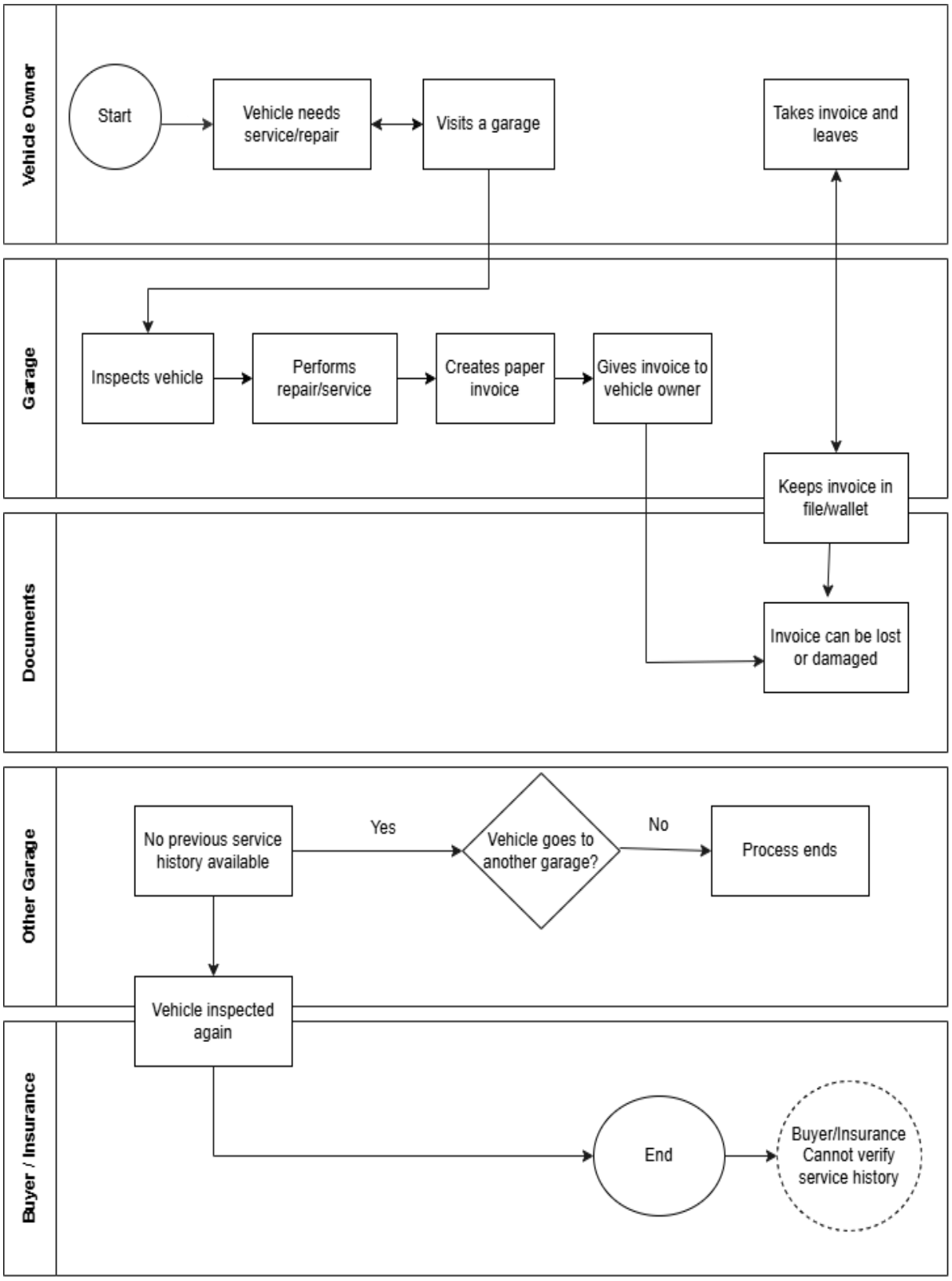
10. Business Rules

Rule ID	Rule
BRULE-01	Only verified garages can update service records
BRULE-02	Owners must approve third-party access
BRULE-03	Shared links must have expiry dates
BRULE-04	Every record update must be logged
BRULE-05	Vehicle records belong to the vehicle owner

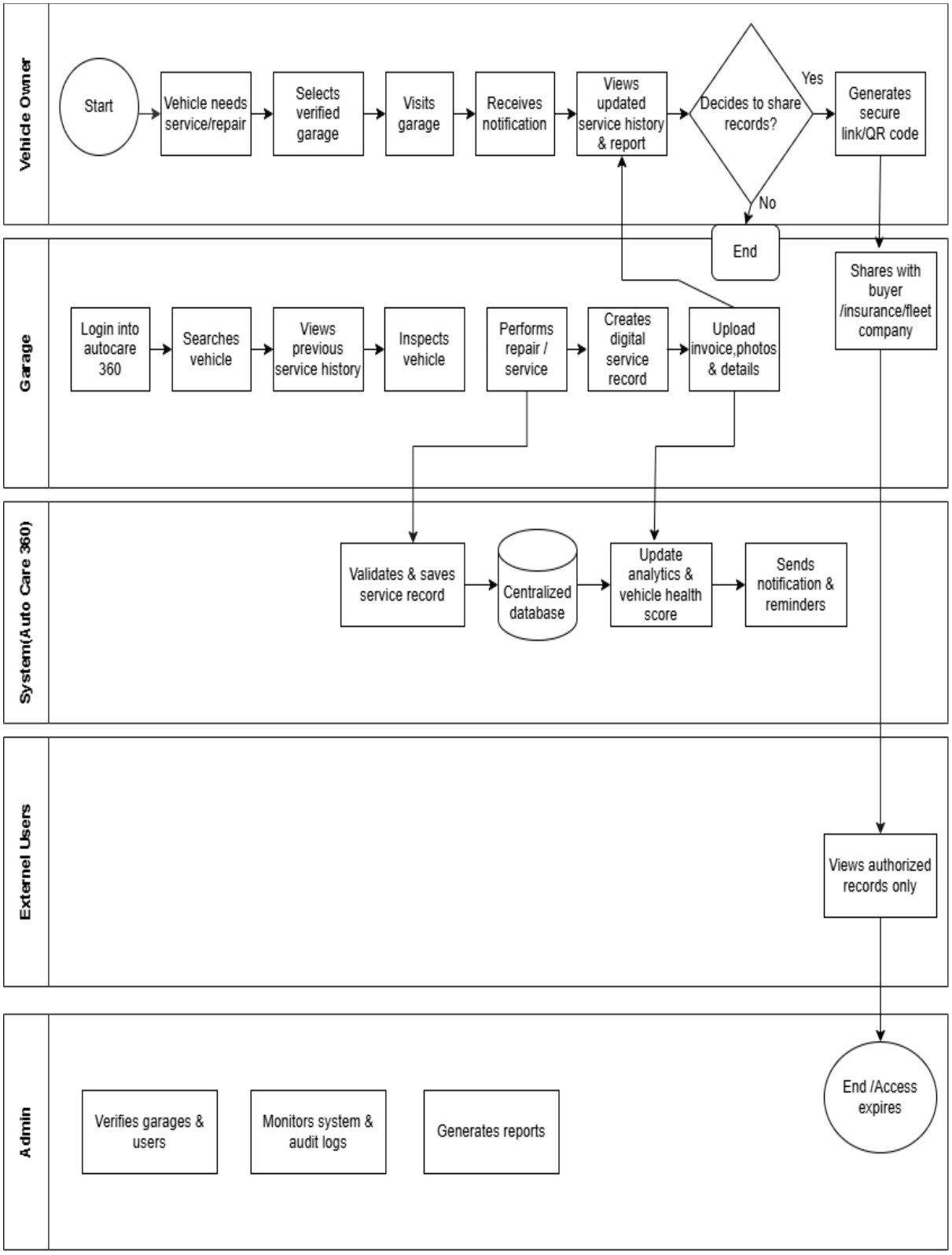
11.Proposed Technology Solution

Frontend	• Flutter Mobile Application • React Web Application
Backend	• Microsoft Azure • Azure App Service • Azure SQL Database
AI Services	• Microsoft Copilot • OCR Processing • Predictive Analytics

12.Current Process (AS-IS)



13.Proposed Solution (TO-BE)



14.Business Model Canvas

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
• Vehicle garages • Insurance companies • Leasing companies • Fleet management companies • Government vehicle registration authority • Cloud service provider (Microsoft Azure) • AI service provider	• Register vehicles • Manage maintenance records • Verify garages • Generate reminders • Secure record sharing • AI maintenance prediction • Store digital documents • Generate reports and dashboards	• One digital maintenance record for each vehicle • Secure and centralized service history • Owner controls data sharing • Easy vehicle resale verification • AI maintenance recommendations • Reduced paperwork and fraud	• Mobile app support • Web portal support • Customer support • Email & SMS notifications • AI chatbot assistance • User feedback and help center	• Vehicle owners • Garages • Fleet operators • Insurance companies • Leasing companies • Vehicle buyers • Government inspectors
	Key Resources		Channels	
	• AutoCare360 platform • Mobile application • Web portal • Azure cloud infrastructure • SQL database • AI engine • Development team • Customer support team		• Android App • iOS App • Web Portal • Email • SMS Notifications • QR Code Sharing • Secure Share Links	

Cost Structure	Revenue Streams
• System development cost • Cloud hosting (Azure) • Database maintenance • AI service cost • SMS & email notification cost • System maintenance and upgrades	• Monthly garage subscription • Premium vehicle owner subscription

• Customer support • Security and backup services	• Fleet management package • Premium analytics package • AI vehicle health reports • Enterprise solutions for insurance and leasing companies